

Aaron Yan

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EDUCATION

University of California, Berkeley

B.A. in Molecular and Cellular Biology | *College of Letters and Sciences*

B.A. in Data Science | *College of Computing, Data Science, and Society*

Honors: Dean's List

May, 2026

GPA: 3.96

SKILLS

- **Programming**: Python, R, Java, SQL, Bash, Git, MATLAB
- **ML / Data**: PyTorch, scikit-learn, NumPy, pandas
- **Bioinformatics & Genomics**: DESeq2, edgeR, limma/voom; FastQC; STAR, Salmon/Kallisto; Scanpy/Seurat; ComBat/pyComBat; GEO; Epi2Me, SLURM/HPC
- **Imaging & Structural**: Imaris, ImageJ/Fiji, PyMOL, FATCAT, MUSCLE
- **Wet-lab**: IHC/IF, microscopy, qPCR, Western blot, flow cytometry, cell culture, animal handling, genotyping (PCR + gel)

PROFESSIONAL EXPERIENCE

UC Berkeley Flanagan Lab

Undergraduate Researcher

Jan 2024 – Present

Berkeley, CA

- Analyzed 5 bulk RNA-seq datasets: PCA, Differential Expression, and Gene Ontology/KEGG Enrichment
- Whole-mounted 40+ mouse retinas and traced 150+ astrocytes in Imaris to compute morphology statistics

UC Berkeley Nogales Lab

Undergraduate Researcher

Jan 2025 – Present

Berkeley, CA

- Analyzed 4 PDB proteins: performed structural superpositions in PyMOL and structure/sequence alignments with FATCAT and MUSCLE to verify proposed evolutionary relationships

UC Berkeley DNA Sequencing Facility

Undergraduate Researcher

Aug 2024 – Present

Berkeley, CA

- **Nanopore**: Ran pre-written Epi2Me + HPC workflows for 2,000+ samples; debugged pipeline issues
- **Sanger**: Set up PCRs for 5,000+ samples for sequencing

Mt. Sinai Medical Center

Bioinformatics Researcher

Jun 2025 – Aug 2025

New York, NY

- Applied and evaluated batch-correction workflows to align pseudo-bulk scRNA-seq with bulk RNA-seq, improving cross dataset concordance across 7 curated datasets
- Queried NCBI GEO and screened 500+ studies to identify the 7 usable datasets

New York Blood Center

Undergraduate Researcher

Jun 2024 – Aug 2024

New York, NY

- Ran a weekly murine tissue workflow (dissection → prep → microscopy / flow cytometry / Western blot/colony culture) across two projects; processed 14 mice total over 2 months

Gladstone Institutes

Undergraduate Researcher

Jun 2023 – Dec 2023

San Francisco, CA

- Genotyped 500+ mice (DNA extraction → PCR → gel electrophoresis) for Nature (2024) study on fibrin-driven thromboinflammation

SELECTED PUBLICATIONS

- Ryu JK, ..., Yan, ... 2024. Fibrin drives thromboinflammation and neuropathology in COVID-19. *Nature*. (co-author)

SELECTED PROJECTS

ML Project (PathMNIST — PyTorch)

- Built CNN classifiers and a U-Net segmentation pipeline in PyTorch; implemented training/validation loops and ran ablations across architectures; achieved ≥96% acc and ≥98% IoU on PathMNIST

Software Dev Project (Java — 2D World Engine)

- Shipped a Java 2D tile-based world exploration engine with procedural world generation, main menu/HUD, and save/load (serialization)